

Penetration Testing Course1

Rubric 1

Network Testing

Engagement title: Home Lab Internal Network Assessment **Phase:** PTES Phase 2–4 (Enumeration, Mapping & Protocol Analysis) **Document version:** 1.0 **Classification:** Confidential — Lab Use Only **Prepared by:** *[Your name]*, DAE Cybersecurity Program **Date:** *[Fill in]* **Related documents:** Engagement Plan (R1), Reconnaissance Report (R2), Vulnerability Assessment (R3)

Same engagement, same target (*192.168.78.5*), same authorization as the previous three rubrics. This report focuses on network protocols, service enumeration, mapping, and live traffic analysis. Placeholders in *[brackets]* are yours to fill.

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1. Objectives & Scope

Objectives:

1. Demonstrate understanding of network protocols through hands-on testing.
2. Enumerate all identified services and document methods and findings.
3. Map the network and visualize its structure and relationships.
4. Identify and analyze access points and their security implications.
5. Perform live traffic analysis and produce findings and recommendations.

Scope (carried forward from R1): 192.168.78.5 and the 192.168.78.0/24 lab segment only.

2. Network Protocols Under Test

This section demonstrates practical protocol knowledge. For each protocol, state what it does, which OSI layer it lives at, and how you tested it.

Protocol	Layer	Purpose	How you tested it	What you observed
ARP	2 (Data Link)	Maps IP addresses to MAC addresses on the local network	<code>arp -a</code> ; watch <code>arp</code> packets in Wireshark	The target's MAC address, proving it's on the local segment
ICMP	3 (Network)	Reachability testing (ping)	<code>ping 192.168.78.5</code>	Replies confirm the host is up
TCP	4 (Transport)	Reliable connection setup via the 3-way handshake	Filter <code>tcp.flags.syn==1</code> in Wireshark during a connection	SYN → SYN-ACK → ACK before any data
Telnet	7 (Application)	Remote terminal access — unencrypted	Capture a login (Section 6)	Credentials readable in plaintext
SSH	7 (Application)	Remote terminal access — encrypted	Capture an SSH login for contrast	Payload is unreadable ciphertext

The key demonstration: capture a telnet login and an SSH login side by side. In Wireshark you can read the telnet session word for word, but the SSH session is scrambled. That single comparison proves you understand the difference encryption makes at the protocol level — and it's the backbone of your findings.

3. Network Mapping & Visualization

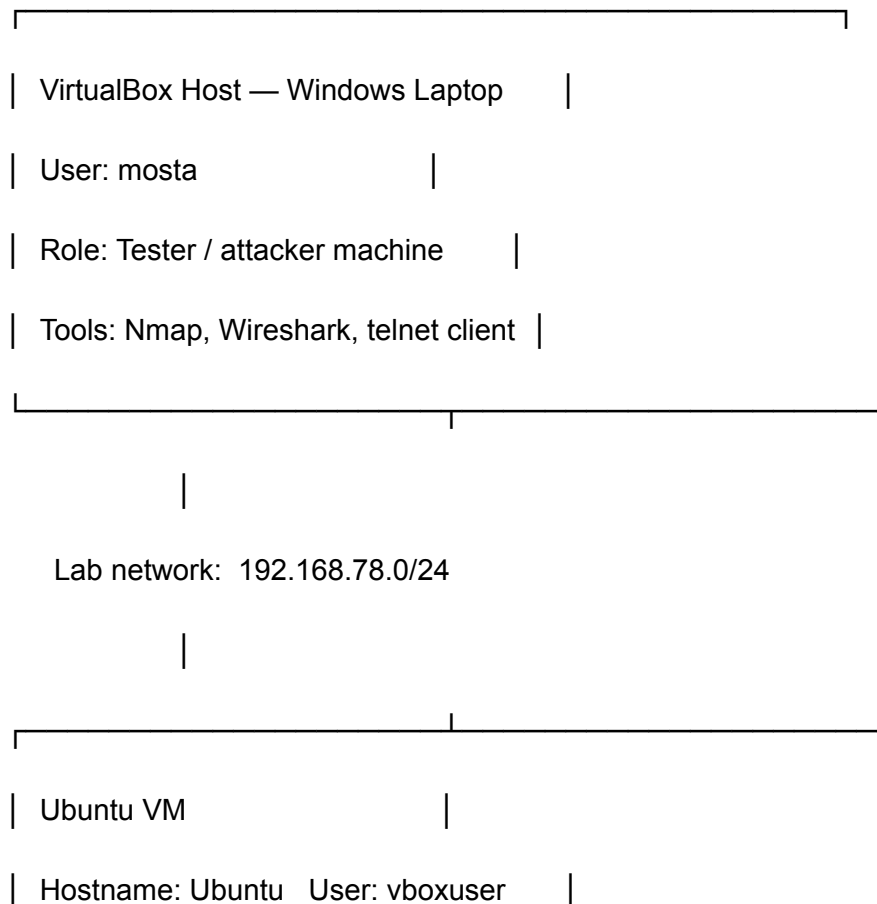
3.1 Discovery method

```
nmap -sn 192.168.78.0/24 -oN network-map.txt # live hosts
```

```
arp -a # IP-to-MAC on the segment (Windows or Linux)
```

```
ip route # identify the gateway (on Ubuntu)
```

3.2 Network diagram



IP: 192.168.78.5	
Role: Target system	
Open services: 23/tcp telnet [22/tcp ssh]	

3.3 Tool-generated visualization (evidence)

Open your all-ports scan in **Zenmap** (the Nmap GUI) and click the **Topology** tab. It draws the discovered hosts and their relationships automatically. Screenshot it — a tool-generated topology map is strong portfolio evidence and satisfies "clear visualization."

3.4 Host relationship table

Host	IP	MAC (from arp)	Role	Relationship
Windows laptop	<i>[your IP]</i>	<i>[MAC]</i>	Tester	Same L2 segment as target
Ubuntu VM	192.168.78.5	<i>[MAC]</i>	Target	Reachable directly from tester
Gateway	<i>[from ip route]</i>	<i>[MAC]</i>	Router	Default route for the segment

4. Service Enumeration

Enumeration means going service-by-service to extract every detail each one reveals. Document the method **and** the finding for each.

4.1 Methodology

1. Identify open ports (from the R2 scans).
2. Grab each service's banner to confirm the software and version.
3. Run protocol-specific Nmap scripts for deeper detail.

4.2 Service: Telnet (23/tcp)

Banner + version

```
sudo nmap -sV -p23 192.168.78.5
```

Telnet-specific script: does it support encryption?

```
sudo nmap -p23 --script telnet-encryption 192.168.78.5
```

Manual banner grab

```
telnet 192.168.78.5 23
```

Findings: service confirmed as *[telnetd version]*; the `telnet-encryption` script reports the service does **not** support encryption — confirming the cleartext weakness at the protocol level.

4.3 Service: SSH (22/tcp) — if present

```
sudo nmap -sV -p22 192.168.78.5
```

```
sudo nmap -p22 --script ssh2-enum-algos,ssh-hostkey 192.168.78.5
```

Findings: *[OpenSSH version, host key fingerprint, supported algorithms]*. Note whether any weak/legacy algorithms are offered.

4.4 Enumeration summary

Port	Service	Version	Enumeration method	Key finding
23/tcp	telnet	[]	-sV, <code>telnet-encryption</code> , banner grab	No encryption supported
[22/tcp]	[ssh]	[]	-sV, <code>ssh2-enum-algos</code>	[]

5. Access Point Identification & Analysis

"Access points" = the ways someone could get into the environment or the target. Identify each, and analyze what it exposes.

Access point	Type	How it grants access	Security implication
Telnet 23/tcp	Remote admin service	Interactive shell after login	High — cleartext credentials can be sniffed; leads to admin compromise
[SSH 22/tcp]	Remote admin service	Encrypted shell after login	Lower — encrypted, but still a login surface; depends on credential strength
Lab network segment	Network entry	Physical/logical access to 192.168.78.0/24 puts an attacker in sniffing range	Flat network with no segmentation amplifies the telnet risk

Analysis: the most dangerous access point is Telnet, because reaching it requires only presence on the local segment and its traffic is readable. Combined with an un-segmented network, one foothold on the segment is enough to capture admin credentials and take over the target.

6. Traffic Analysis

This is the practical heart of the rubric — and the solution to the capture you've been troubleshooting.

6.1 Reliable capture setup

You have two places to capture. The second is foolproof.

Option A — Wireshark on the Windows laptop:

- On the start screen, watch the mini traffic graphs while you `ping 192.168.78.5`. Capture on **whichever interface's graph moves** — with VirtualBox this is often the "VirtualBox Host-Only Network" adapter, *not* your Wi-Fi.
- Start the capture **before** logging in, apply display filter `tcp.port == 23`, then run the telnet login.

Option B — tcpdump on the Ubuntu VM (guaranteed — use this if A gives nothing):

```
sudo tcpdump -i any -A 'tcp port 23' -w telnet-capture.pcap
```

`-i any` removes the wrong-interface problem entirely; `-A` shows text live; `-w` saves the file. Log in from the laptop, then `Ctrl+C`. Open `telnet-capture.pcap` in Wireshark for the writeup.

Why your earlier captures showed nothing: almost always the wrong interface, or the capture started after login. Option B sidesteps both.

6.2 Protocol analysis walkthrough

1. **TCP handshake** — filter `tcp.flags.syn==1`. You'll see SYN, SYN-ACK, ACK: the connection being established before any data moves.
2. **Cleartext credentials** — filter `telnet`, right-click a packet → **Follow** → **TCP Stream**. The username and password appear as readable text. This is the flagship evidence — screenshot it.
3. **Encryption contrast** — capture an SSH login the same way (`tcp.port == 22`, Follow TCP Stream). The content is unreadable ciphertext. Screenshot both windows together.

6.3 Findings

#	Observation	Evidence
T-01	Telnet credentials transmitted in cleartext and fully recoverable	<code>telnet-capture.pcap</code> , TCP stream screenshot
T-02	TCP handshake and session visible; no transport encryption on port 23	Handshake screenshot
T-03	<i>[SSH session unreadable — encryption working]</i>	<i>[SSH stream screenshot]</i>

6.4 Recommendations

- Replace Telnet with SSH so session data (including credentials) is encrypted.
 - Segment the lab network so hosts can't freely sniff each other's traffic.
 - Consider monitoring for plaintext-protocol usage as a detection control.
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7. Findings Summary & Recommendations

ID	Finding	Severity	Recommendation
T-01	Cleartext credentials over Telnet	High	Disable Telnet; use SSH
T-03	No network segmentation	Medium	Isolate segments / restrict sniffing scope
[]	[]	[]	[]

These are consistent with the R3 vulnerability assessment — this report adds the protocol-level and traffic-level proof behind them.

8. Evidence Log

- ☐ `network-map.txt` — host discovery output
 - ☐ Zenmap Topology screenshot
 - ☐ `arp -a` output (IP-to-MAC mapping)
 - ☐ Service enumeration output (`-sV`, `telnet-encryption`)
 - ☐ `telnet-capture.pcap` — telnet traffic capture
 - ☐ Wireshark **Follow TCP Stream** screenshot showing plaintext credentials
 - ☐ TCP handshake screenshot (`tcp.flags.syn==1`)
 - ☐ (If SSH set up) SSH stream screenshot showing encrypted payload
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Appendix A — Wireshark Filter Reference

Goal	Display filter
Only telnet traffic	<code>tcp.port == 23 or telnet</code>
Only the target	<code>ip.addr == 192.168.78.5</code>
TCP handshake packets	<code>tcp.flags.syn == 1</code>
ARP traffic	<code>arp</code>
Ping traffic	<code>icmp</code>
SSH traffic	<code>tcp.port == 22</code>

Right-click any packet → **Follow** → **TCP Stream** to reassemble a full conversation.

Appendix B — Command Reference & References

Goal	Command
Map live hosts	<code>nmap -sn 192.168.78.0/24 -oN network-map.txt</code>
IP-to-MAC table	<code>arp -a</code>
Enumerate telnet	<code>sudo nmap -sV -p23 --script telnet-encryption 192.168.78.5</code>
Capture on target	<code>sudo tcpdump -i any -A 'tcp port 23' -w telnet-capture.pcap</code>
Identify gateway	<code>ip route</code>

- NIST SP 800-115, *Technical Guide to Information Security Testing and Assessment*
- Nmap NSE documentation — nmap.org/nsedoc
- Wireshark User's Guide — [wireshark.org/docs](https://www.wireshark.org/docs)
- TCP/IP Illustrated (Stevens) — protocol reference

This report is for authorized security testing in a controlled lab environment only.